Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Naming Plane Shapes and Their Properties

Plane shapes are also known as two-dimensional (2-D) shapes. They lie flat on a surface, and the shapes you will meet are the triangle, the rectangle, the square, the parallelogram, the trapezium, the circle and other polygons.

A polygon is a closed shape that has three or more straight sides. Polygons are named by the number of sides they have. The table below lists the common ones, together with the number of triangles you can divide each polygon into by drawing diagonals from one vertex to all the non-adjacent vertices.

Name of polygon	Number of sides	Number of triangles
Triangle	3	1
Quadrilateral	4	2
Pentagon	5	3
Hexagon	6	4
Heptagon	7	5
Octagon	8	6
Nonagon	9	7
Decagon	10	8
Hendecagon (or undecagon)	11	9
Dodecagon	12	10

Notice the pattern: the number of triangles is always two fewer than the number of sides, because you subtract 2 from the number of sides. So an octagon, with 8 sides, divides into $8 - 2 = 6$ triangles; a pentagon, with 5 sides, divides into $5 - 2 = 3$ triangles.

Now look at the three kinds of triangles in fig-01. The tick marks on the sides show which sides are equal (a single tick means "equal to the other single tick", and one, two or three ticks show three different lengths), and the arcs at the corners mark the angles.

An isosceles triangle has two adjacent sides that are equal, its two base angles are equal, and it has one line of symmetry.

A scalene triangle has no side equal to another, no angle equal to another, and no line of symmetry.

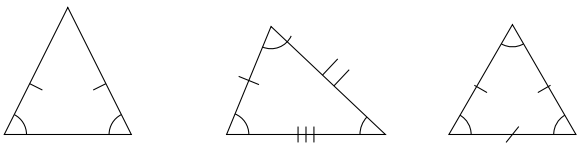
An equilateral triangle has all three sides equal, all three angles equal (each measuring 60°), and three lines of symmetry.

Here is how they compare:

Isosceles	Scalene	Equilateral
2 adjacent sides are equal	No side is equal	3 sides are equal
Base angles are equal	No angle is equal to another	All angles are equal (60° each)
One line of symmetry	No line of symmetry	Three lines of symmetry

The main difference between them is how many sides are equal: an equilateral has all three equal, an isosceles has just two equal, and a scalene has none equal. That is why the equilateral has three lines of symmetry, the isosceles has one, and the scalene has none.

Visual



Isosceles Scalene Equilateral

Isosceles: a single tick on each of the two slanted sides, an arc at each base angle. Scalene: one, two and three ticks on the three sides, an arc at each vertex. Equilateral: a single tick on each of the three sides, an arc at each vertex.

fig-02 shows a square and a rectangle side by side. In both, a small square mark at each corner marks a right angle (90°), and the tick marks on the sides show which sides are equal.

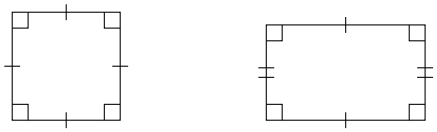
A square has all four sides equal, two pairs of parallel sides, all four angles equal (each 90°), four lines of symmetry, and its diagonals bisect each other at a right angle.

A rectangle has two opposite sides equal (the two longer sides match, and the two shorter sides match), two pairs of parallel sides, all four angles equal (each 90°), two lines of symmetry, and its diagonals bisect each other but not at a right angle.

Square	**Rectangle**
All sides are equal	2 opposite sides are equal
2 pairs of parallel sides	2 pairs of parallel sides
All angles are equal (90° each)	All angles are equal (90° each)
Four lines of symmetry	Two lines of symmetry
Diagonals bisect it at a right angle	Diagonals bisect it but not at a right angle

The two shapes agree on parallel sides and right angles. They differ on the sides (a square has all four equal, a rectangle only opposite ones equal), on the number of lines of symmetry (4 for a square, 2 for a rectangle), and on whether the diagonals cut at a right angle.

Visual



Squares Rectangles

Square: a right-angle mark at each corner, a single tick on each of the four sides. Rectangle: a right-angle mark at each corner, a single tick on each of the two longer sides and a double tick on each of the two shorter sides.

fig-03 shows a parallelogram and a trapezium. The arrows on the sides mark the sides that are parallel (a single arrow on two sides means those two sides are parallel to each other, and a double arrow marks the other pair).

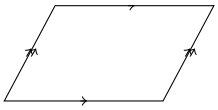
A parallelogram has two pairs of opposite sides that are equal and parallel, its opposite angles are equal, it has no line of symmetry, and its diagonals divide it into two equal triangles.

A trapezium has only one pair of parallel sides, its opposite angles are not equal, it has no line of symmetry, and its diagonal divides it into two unequal triangles.

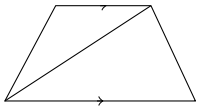
Parallelogram	**Trapezium**
2 pairs of opposite sides are equal and parallel	Only a pair of sides are parallel
Opposite angles are equal	Opposite angles are not equal
No line of symmetry	No line of symmetry
Diagonals divide it into equal triangles	Diagonal divides it into two unequal triangles

The key difference is the number of parallel sides: the parallelogram has two pairs of parallel sides, while the trapezium has only one pair. Because of that, the parallelogram's opposite angles are equal and its diagonals make equal triangles, while the trapezium's are not equal and do not.

Visual



Parallelogram



Trapezium

Parallelogram: a single arrow on each of the two horizontal sides, a double arrow on each of the two slanted sides. Trapezium: a single arrow on each of the two parallel sides, and one diagonal drawn in.

Practice

1. A shape has four sides, with only one pair of them parallel. Which shape is it, and how many lines of symmetry does it have?
2. A triangle has no two sides equal. What is this kind of triangle called, and what does that tell you about its angles?
3. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?
4. A square and a rectangle both have two pairs of parallel sides and all angles equal to 90°. State one difference between a square and a rectangle.
5. What is the name of the polygon with 12 sides, and how many triangles does it divide into?

Section 2: The Parts of a Circle

A circle is a plane shape drawn so that every point on its edge is the same distance from one fixed point inside it, called its **centre**. Like every plane shape, a circle has parts, and being able to name them is the first step to working with circles. Look at fig-04, where two circles are drawn with their parts labelled.

The **circumference** is the whole curved line that forms the circle itself — its outer boundary. Imagine running a pencil all the way around the circle without lifting it; the line you trace is the circumference.

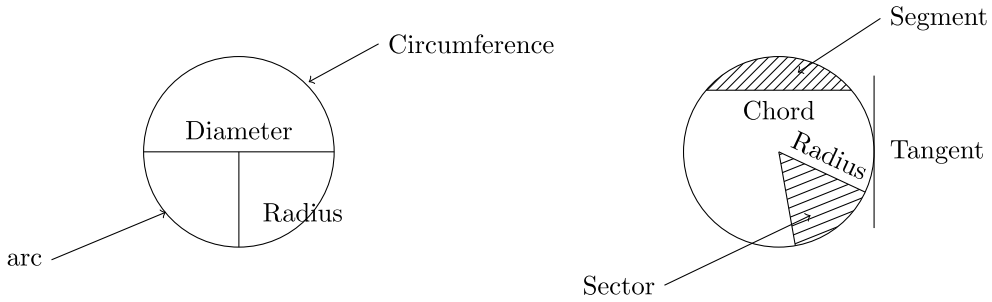
A small curved part of the circumference is called an **arc**. If you draw a circle and only trace a portion of its curved edge, that portion is an arc. The first circle in fig-04 shows an arc marked along part of its circumference.

The **radius** is any straight line that joins the centre of the circle to the circumference. It is the distance from the centre out to the edge. In fig-04, the first circle has a radius drawn from its centre down to the circumference, and the second circle shows one of its sector's edges labelled Radius too. Every circle has many radii, and they are all the same length — that is what "every point on the edge is the same distance from the centre" means.

The **chord** is any straight line joining two points on the circumference. It runs across the inside of the circle from one point on the edge to another. The second circle in fig-04 shows a chord drawn near the top.

The **diameter** is a special chord: any chord that passes through the centre of the circle. Because it goes through the centre, the diameter is the longest straight line you can draw across a circle, and it is exactly twice as long as a radius. In fig-04 the first circle has its diameter drawn straight across through the centre.

Visual



Two labelled circles. The first has the diameter drawn across it and a radius from its centre to the circumference, with the arc and the circumference labelled. The second has a chord near the top with the segment above it shaded, a shaded sector below with its right-hand bounding line labelled Radius, and a tangent touching it on the right.

Two more parts of a circle are named from the regions the lines cut out of it.

A **segment** is the region of the circle cut off by a chord. In the second circle of fig-04, the chord near the top cuts the circle, and the shaded part above the chord is a segment.

A **sector** is a region of the circle shaped like a slice, bounded by two radii and the arc between them. In fig-04, the second circle shows a shaded sector with two radii as its straight sides and an arc between them. The right-hand side of that sector is labelled Radius.

Finally, the **tangent** is a straight line that touches the circumference at exactly one point and does not cut through the circle. In fig-04, the line on the right-hand side of the second circle touches the circle at a single point — that is a tangent.

Practice

1. Name the part of the circle that is the whole curved line forming the circle's outer boundary.
2. A straight line joins two points on the circumference and passes through the centre of the circle. What is this line called?
3. A straight line touches the circumference at exactly one point without cutting through the circle. What is this line called?
4. A straight line joining the centre of a circle to the circumference is called a radius. If the radius of a circle is 4 cm, how long is its diameter?

Section 3: The Perimeter of Plane Shapes

The **perimeter** of a shape is the total distance all the way round it. To find it you add up the lengths of every side that makes the boundary of the shape.

For shapes with sides of equal length, you can multiply. A **regular polygon** is a shape whose sides are all the same length, so its perimeter is one side length multiplied by the number of sides.

Here are the forms you will use:

- **Square** — all four sides are equal. Perimeter $= L + L + L + L = 4L$.
- **Rectangle** — two long sides (length L) and two short sides (breadth b). Perimeter $= L + b + L + b = 2L + 2b$.
- **Trapezium** — four sides, all possibly different. Perimeter $= a + b + c + d$.
- **Parallelogram** — two pairs of equal opposite sides. Perimeter $= 2a + 2b$.
- **Regular polygon** — n equal sides each of length s. Perimeter $= n \times s$.
- **Circle** — its perimeter is called the **circumference**. Circumference $= 2\pi r = \pi d$, where r is the radius and d is the diameter, with $d = 2r$.

Because perimeter is a length, every answer is written with a unit such as cm.

Worked example 1 — square. Find the perimeter of a square of side 15 cm.

Each side is $L = 15$ cm and there are four of them:
$$P = 4L = 4 \times 15 = 60 \text{ cm}.$$

Worked example 2 — rectangle. Find the perimeter of a rectangle of length 10 cm and breadth 7 cm.

$$P = 2L + 2b = 2(10) + 2(7) = 20 + 14 = 34 \text{ cm}.$$
Check: $10 + 7 + 10 + 7 = 34$ cm, the same answer.

Worked example 3 — rectangle, finding a missing side. If the perimeter of a rectangle is 60 cm and its length is 17 cm, find its breadth.

Write the perimeter formula and substitute what you know:
$$60 = 2(17) + 2b = 34 + 2b.$$
Collect the like terms:
$$60 - 34 = 2b \quad \Rightarrow \quad 2b = 26.$$
Divide both sides by 2:
$$b = \frac{26}{2} = 13 \text{ cm}.$$

Worked example 4 — trapezium. Find the perimeter of a trapezium with sides $a = 11$ cm, $b = 9$ cm, $c = 5$ cm and $d = 4$ cm.

Add all four sides:
$$P = a + b + c + d = 11 + 9 + 5 + 4 = 29 \text{ cm}.$$

Worked example 5 — parallelogram. Find the perimeter of a parallelogram with base $a = 9$ cm and slanted side $b = 5$ cm.

The two bases are equal and the two slanted sides are equal:
$$P = 2a + 2b = 2(9) + 2(5) = 18 + 10 = 28 \text{ cm}.$$

Worked example 6 — regular hexagon. Find the perimeter of a regular hexagon of side 5.2 cm.

A hexagon has 6 sides, all equal:
$$P = 6 \times 5.2 = 31.2 \text{ cm}.$$

Worked example 7 — circle. Calculate the perimeter (circumference) of a circle of radius 7 cm. Use $\pi = \frac{22}{7}$.

Using the radius:
$$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 2 \times 22 = 44 \text{ cm}.$$
The same result comes from πd: since $d = 2r = 14$ cm,
$$C = \frac{22}{7} \times 14 = 22 \times 2 = 44 \text{ cm}.$$

Worked example 8 — circumference, finding the radius. The circumference of a circle is 440 cm. Find the radius, using $\pi = \frac{22}{7}$.

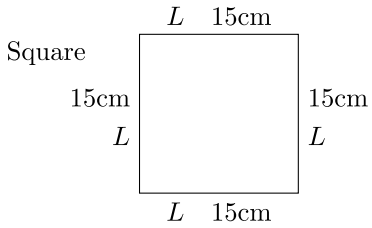
\[440 = 2\pi r = 2 \times \frac{22}{7} \times r = \frac{44r}{7}.$$ Cross-multiply: $$440 \times 7 = 44r \quad\Rightarrow\quad 3080 = 44r.$$ Divide both sides by 44: $$r = \frac{3080}{44} = 70\ \text{cm}.\]

Worked example 9 — a shape made from a quarter circle. A shape is a quarter circle: two straight sides of 4.9 cm meeting at a right angle, closed by an arc. Find its perimeter.

First find the length of the curved part. It is a quarter of a full circumference with $r = 4.9$ cm:
$$\frac{1}{4} \times 2\pi r = \frac{1}{4} \times 2 \times \frac{22}{7} \times 4.9.$$
Since $2 \times \frac{22}{7} \times 4.9 = \frac{44}{7} \times \frac{49}{10} = \frac{44 \times 7}{10} = 30.8$, a quarter of this is $30.8 \div 4 = 7.7$ cm.

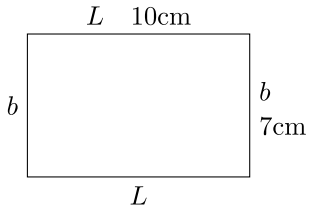
Add the two straight sides and the arc:
$$P = 4.9 + 4.9 + 7.7 = 17.5 \text{ cm}.$$

Visual



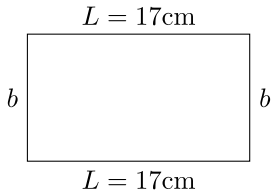
A square, named *Square* at its left, with all four sides marked twice — L and 15cm.

Visual



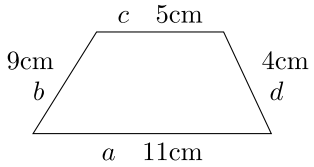
A rectangle with both long sides marked L, the top one also 10cm, and both short sides marked b, the right-hand one also 7cm.

Visual



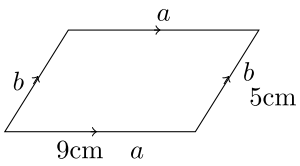
A rectangle with both long sides marked $L = 17$cm and both short sides marked b.

Visual



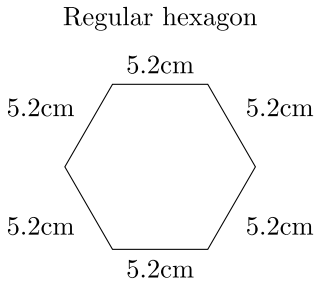
A trapezium with each side lettered and measured: the top c, 5cm; the left b, 9cm; the right d, 4cm; the base a, 11cm.

Visual



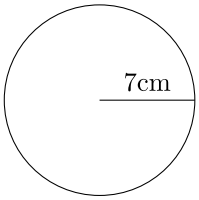
A parallelogram with a single arrow on each of its four sides: the two horizontal ones both lettered a, the lower also measured 9cm, and the two slanted ones both lettered b, the right-hand one also measured 5cm.

Visual



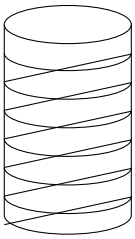
A regular hexagon, named above the drawing, with each of its six sides marked 5.2cm.

Visual



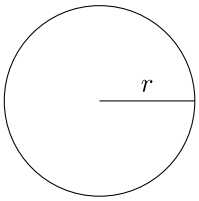
A circle with a radius drawn from the centre to the right of the circumference, marked 7cm.

Visual



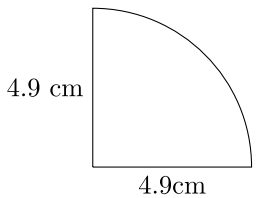
An upright cylinder with a rope wound round it, drawn as a series of turns spiralling down the body from the top face to the base.

Visual



A circle with a radius drawn from the centre to the right of the circumference, marked r.

Visual



A quarter circle: an upright side and a base meeting at a right angle, each marked 4.9cm, with an arc closing them from the top of the upright to the end of the base.

Practice

1. Find the perimeter of a square of side 8 cm.
2. Find the perimeter of a rectangle of length 7 cm and breadth 4 cm.
3. What is the perimeter of a regular pentagon with sides of 5 cm?
4. Find the perimeter of a parallelogram whose base is 6 cm and slanted side is 5 cm.

5. The circumference of a circle is 22 cm. Find its radius, using $\pi = \frac{22}{7}$.
6. A rectangle is such that its length is twice its breadth. If the breadth is d, write an expression for the total length of all the sides of the rectangle.
7. Find the perimeter of a regular hexagon with sides of 5 cm.

Check your work

Attempt every question before you read this page.

Naming Plane Shapes and Their Properties

1. It is a trapezium, and it has no line of symmetry.

   A trapezium is the shape whose only special feature is that it has a single pair of parallel sides. Its properties list shows that it has no line of symmetry.
2. It is a scalene triangle, and no angle is equal to another.

   A triangle with no equal sides is scalene. Its properties are: no side is equal, no angle is equal to another, and it has no line of symmetry.
3. 6 triangles.

   The number of triangles is two fewer than the number of sides. An octagon has 8 sides, so it divides into 8 − 2 = 6 triangles.
4. A square has all four sides equal, while a rectangle has only opposite sides equal (or: a square has four lines of symmetry, a rectangle has two).

   Compare the two rows of the square/rectangle table. The square has all sides equal and four lines of symmetry; the rectangle has opposite sides equal and two lines of symmetry. Either difference is correct.
5. It is a dodecagon, and it divides into 10 triangles.

   From the polygon table, a 12-sided shape is a dodecagon. Its number of triangles is 12 − 2 = 10.

The Parts of a Circle

1. The circumference.
2. The diameter (a diameter is any chord that passes through the centre).
3. A tangent.
4. 8 cm, because the diameter is twice the radius.

   The diameter is any chord that passes through the centre, and it is twice as long as a radius. So diameter = 2 × radius = 2 × 4 cm = 8 cm.

The Perimeter of Plane Shapes

1. 32 cm

   $P = 4L = 4 \times 8 = 32$ cm.
2. 22 cm

   $P = 2L + 2b = 2(7) + 2(4) = 14 + 8 = 22$ cm.
3. 25 cm

   A pentagon has 5 equal sides, so $P = 5 \times 5 = 25$ cm.
4. 22 cm

   $P = 2a + 2b = 2(6) + 2(5) = 12 + 10 = 22$ cm.
5. 3.5 cm

   $22 = 2\pi r = 2 \times \frac{22}{7} \times r = \frac{44r}{7}$. Cross-multiply: $22 \times 7 = 44r$, so $154 = 44r$ and $r = \frac{154}{44} = 3.5$ cm.
6. $6d$

   Length $L = 2d$ and breadth $b = d$. Perimeter $= 2L + 2b = 2(2d) + 2(d) = 4d + 2d = 6d$.
7. 30 cm

   A hexagon has 6 equal sides, so $P = 6 \times 5 = 30$ cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-13d32c0f	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____